

Chapter 2. PERMUTATIONS

2.3 The Alternating group



In this video we will prove the **PARITY THEOREM**.

Proving the **Parity Theorem**.

THEOREM 2.20. **PARITY THEOREM.**

If a permutation σ has two factorisations

$$\sigma = \gamma_r \dots \gamma_2 \gamma_1 = \mu_s \dots \mu_2 \mu_1,$$

where each γ_i and μ_j is a transposition, then both r and s are even or both are odd.

We first need to prove a preliminary result:

LEMMA 2.22. The identity permutation ε can not be written as the product of an odd number of transpositions.

Proof. Suppose that $\varepsilon = \gamma_1 \gamma_2 \dots \gamma_r$ is a product of transpositions.

Goal: To show that r is even.

We proceed by induction on $r \geq 2$.

$r=2$. There is nothing prove.

Suppose that $r > 2$ and the result is true for any factorisation of the identity permutation E as a product of $< r$ transpositions.

$$E = \tau_1 \tau_2 \dots \underbrace{\tau_{r-1} \tau_r}$$

We first look at τ_{r-1} and τ_r .

If $\tau_r = (a \ b)$ (with $a \neq b$), then there are four possibilities for τ_{r-1} :

$$\textcircled{1} \tau_{r-1} = (b \ a) = (a \ b) = \tau_r$$

In this case $\tau_{r-1} \tau_r = E$, and so

$E = \tau_1 \tau_2 \dots \tau_{r-2}$ is a product of $r-2$ transpositions. By the inductive hypothesis we obtain that $r-2$ is even, and so r is even.

② $\tau_{r-1} = (a \ c)$ for $c \notin \{a, b\}$.

Then $\tau_{r-1} \tau_r = (a \ c)(a \ b)$
 $= (a \ b)(b \ c)$
 a appears a is not here

③ $\tau_{r-1} = (b \ c)$ for $c \notin \{a, b\}$

$\tau_{r-1} \tau_r = (b \ c)(a \ b) = (\underline{a} \ c)(b \ c)$
 a is not here

④ $\tau_{r-1} = (c \ d)$ for $c \neq d$ and $c, d \notin \{a, b\}$

$$\gamma_{r-1} \gamma_r = (c \ d)(a \ b) = \underline{(a \ b)}(c \ d)$$

a is
not here

In any of the cases ②, ③ and ④ we have replaced the product $\gamma_{r-1} \gamma_r$ by the product of other two transpositions $\alpha_{r-1} \alpha_r$ such that a appears in α_{r-1} but not in α_r .

If we continue in this way, at some point we will have to end up with a product of $r-2$ transpositions (and so r is even like in case ①); otherwise we would obtain a product of r transpositions equal to the identity:

$$E = \beta_1 \beta_2 \dots \beta_r$$

such that a appears in β_1 but not
in β_i for $i > 1$. But this is
impossible because $(\beta_1 \beta_2 \dots \beta_r)(a) \neq a$
and so $\beta_1 \beta_2 \dots \beta_r \neq E$, a contradiction. $\blacksquare$

THEOREM 2.20. PARITY THEOREM.

If a permutation σ has two factorisations

$$\sigma = \gamma_r \dots \gamma_2 \gamma_1 = \mu_s \dots \mu_2 \mu_1,$$

where each γ_i and μ_j is a transposition, then both r and s are even or both are odd.

Proof. Suppose that a permutation σ admits two factorisations into transpositions:

$$\sigma = \gamma_n \dots \gamma_2 \gamma_1 = \mu_m \dots \mu_2 \mu_1$$

Goal: To show that both n and m are either even or odd.

Recall that $\mu_j^{-1} = \mu_j$ for all j (they are transpositions).

$$\gamma_n \dots \gamma_2 \gamma_1 = \mu_m \dots \mu_2 \mu_1$$

Multiplying by μ_m on the left we obtain

$$\mu_m \tau_n \dots \tau_2 \tau_1 = \mu_m^2 \mu_{m-1} \dots \mu_2 \mu_1$$

$$\xrightarrow{\mu_m^2 = \epsilon} \mu_m \tau_n \dots \tau_2 \tau_1 = \mu_{m-1} \dots \mu_2 \mu_1$$

$$\Rightarrow \mu_{m-1} \mu_m \tau_n \dots \tau_2 \tau_1 = \mu_{m-1}^2 \mu_{m-2} \dots \mu_2 \mu_1$$

$$\Rightarrow \mu_{m-1} \mu_m \tau_n \dots \tau_2 \tau_1 = \mu_{m-2} \dots \mu_2 \mu_1$$

$$\Rightarrow \dots \Rightarrow \boxed{\mu_1 \mu_2 \dots \mu_m \tau_n \dots \tau_2 \tau_1 = \epsilon}$$

We have obtained a factorisation of the identity permutation as a product of $m+n$ transpositions. By Lemma 2.22, $m+n$ is even.

This implies that either m and n
are even or m and n are odd.

